Literature Review of Explanation using Defeasible Reasoning

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Abstract

Defeasible Reasoning is a form of non-monotonic reasoning that aims to answer why propositions typically hold. However, we may need more information regarding the basic justification for how such reasoning comes to its conclusion. Explanation traces for reasoning systems can benefit the pursuit of interpretability within agent classifiers. Symbolic rule classifiers that encode the default and exception structure directly in the rule bodies are a novel approach in designing predicators that are traceable. Thus, the rule-based classifier provides full traceability: users can see both why a decision applies and why it may be overidden. The aim is to extend existing classifiers by providing a translation tool to KLM style defeasible knowledge bases and then verifying the approach against typically defeasible algorithms such as Rational Closure, Lexicographic Closure and System W. The aim is to produce explanation traces on both the symbolic classifier and the specific entailment algorithms.

Keywords

Artificial Intelligence, Defeasible Reasoning, Propositional Logic, Rule-based Models, KLM Framework, Horn Rules, Transparency, Interpretable Classification, Explainable AI (XAI)

1 Introduction and Motivation

Machine Learning (ML) is being increasingly deployed within environments that have major impact on human life.[**?**] Many of the systems deployed are what is known as 'black-box' algorithms, which implies that they are complex and opaque in nature [**?**]. Black-box algorithms are typically difficult to justify with short, human-readable sentences. On the other hand, algorithms are considered transparent if its internal logic is fully accessible.[**?**] It is important to note that many domains could make use of models that prioritize justification and explanation over specific predictive accuracy. Domains such as healthcare or criminal justice could greatly benefit from transparent algorithms [**?**]. Artificial Intelligence consists of two main approaches, that of machine learning (ML) and that of knowledge representation and reasoning (KRR). The KRR approach uses formal logic to represent knowledge within a domain symbolically. There are many approaches to formal logic with varying degrees of expressiveness. The foundation is that of classical reasoning which describes that any consequence must be maximally supported by explicit knowledge [**?**]. The limitation of this idea is that classical reasoning cannot describe exceptions to their knowledge bases. To be concise we provide a more concrete example. Let us suggest the following:

- Bird → Flies
- Penguin → Bird
- Penguin → ¬Flies

Using classical rules of inferences we have both that **penguin** → **¬flies** and that since **penguin** → **bird** then **penguin** → **flies**. Under classical logic the above statements are a clear contradiction which implies that in no interpretation penguins can exist. This is not to say it is impossible for penguins to exist. An exception could be made by explicit means. The natural implication would mean for any exception of this nature we would need to extend the knowledge base. As one can imagine this will become infeasible if we have to consider every exception to knowledge bases. Defeasible reasoning aims to remedy this by providing a pattern similar to that of human reasoning. We at first can make an assumption about a given statement but upon learning more information we are able to retract any given inference. This is known as defeasible reasoning. What was described regarding absolute inferences in the classical case is based off the property of monotinicity. Defeasible reasoning does not require this property and is classified as non-monotonic [**?**]. The aim of this paper is to provide an overview of the theoretical background of symbolic classifiers such as the Defeasible Horn Classifier with Exceptions (DHCE) [**?**]. Therefore making it possible to build a link between such a classifier and existing defeasible frameworks. In Section 2 we give an overview of propositional logic and classical entailment. In Section 3 we explore the KLM approach. In Section 4 we provide the Rational Closure as well as its extensions. In Section 5 we discuss the current symbolic classifier we will use in our project, ECCRS and its predecessor DHCE. Section 6 and 7 with be a discussion and conclusion with relevance to our approach to traceable symbolic classifiers.

2 Propositional Logic

Propositional Logic is the basic language chosen by Kraus, Lehmann and Magidor when defining the KLM framework [**?**]. First we must introduce the relevant syntax of this logic before we can express the ideas related to defeasible reasoning. A propositional atom is a statement that can be assigned one of two values, conventionally called true and false or T and F.[**?**] Statements are represented by meta variable: $p, q, r, \ldots$ such that b could mean the statement that "birds fly". The set of *atomic propositions* is denoted by P. Formula are created by combining atoms with the set of connectives $\{ \wedge, \vee, \neg, \rightarrow, \leftrightarrow \}$. All connectives are binary except for the unary $\neg$. Each connective will accept all formulas as arguments. With this in mind we can now explore more expressive formulas. The set of all formulas is denoted as $\mathcal{L}$, which has elements $\{\alpha,\beta,\gamma, \ldots\}$. A formula in propositional logic is defined recursively[**?**] such that for any atom in P and any α,β in $\mathcal{L}$ we can formulate: $\alpha : p$, $\neg\ \alpha$, $\alpha \wedge \beta$, $\alpha \vee \beta$, $\alpha \rightarrow \beta$, $\alpha \leftrightarrow \beta$. α is said to be the antecedent and β the consequent. The two constants defined in propositional logic follow that of tautology and contradiction. Denoted by $\top$ and $\bot$ respectively and being elements of the language $\mathcal{L}$. We can now

define the semantics or the meaning of formulas in our language $\mathcal{L}$. Consider the arithmetic expression E such as x + 2 * y, we assign values to x and y and then evaluate the expression. If we let x = 1 and y = 2 we can evaluate the result such that E evaluates to 5. It may be understood that our propositional atoms will only gain meaning when truth values are assigned to the atoms of the formula. Only then can we designate a truth value for the formula as a whole. The definition of an interpretation follows that if we let $A \in P$ be a formula and let P_A be the set of atoms appearing in A. An interpretation for A is a total function $P_A : P_A \mapsto \{ \text{T, F} \}$ that assigns one of the truth values T or F to every atom in P_A. This is known as a valuation but is referred to as worlds or interpretations in the literature [**?**]. If an atom is true in a valuation, then it is said that the valuation satisfies the atom. This is denoted by $\Vdash$ such that for some valuation u, if it is the case that for p $\in P$, $u(p) = T$, then u $\Vdash$ p. Similarly if $u(p) = F$, then u $\nVdash$ p. This can be extended for all α, β, γ,... $\in \mathcal{L}$. We define $\mathcal{U}$ to be the set of all valuations in $\mathcal{L}$. Given the set of valuations we can define further that for any $\alpha \in \mathcal{L}$, let $\hat{\alpha} := \{ u \in \mathcal{U} \mid u \Vdash \alpha \}$. For any $u \in \hat{\alpha}$, u is referred to as a model of α. A formula α is considered satisfiable if there exists a $u \in \mathcal{U}$ such that $u \Vdash \alpha$. A formula α is considered unsatisfiable if there is no such $u \in \mathcal{U}$ such that $u \Vdash \alpha$. dIf it is for all $\alpha \in \mathcal{L}$ then α is only considered a tautology if for all $u \in \mathcal{U}$, it is the case that $u \Vdash \alpha$. Logical consequence is therefore defined by the fact iff $\hat{\alpha} \subset \hat{\beta}$ then for every $u \in \mathcal{U}$ it must be the case that $u \Vdash \alpha$ and $u \Vdash \beta$, or, equivalently, $\alpha \models \beta$, read as β is the logical consequence of α. Finally we can describe the knowledge base $\mathcal{K}$. Let $\mathcal{K}$ be a set of formulas, if $\mathcal{K}$ entails any α if every model of $\mathcal{K}$ is also model of α. $\mathcal{K}$ is generally referred to as a knowledge base and is a finite set of propositional formulas such that $\mathcal{K} \in \mathcal{L}$. This can be understood as a set of facts about the some domain that restrict a set of valuations in $\mathcal{U}$ to satisfy the knowledge base. It should be noted that $\vdash$ as well as $\models$ are meta-level concepts as opposed to object-level symbols such as the connectives mentioned earlier. A meta-level symbol represent what can be said about object-level statements. The idea is that object-level represents knowledge and meta-level represents knowledge about knowledge.

2.1 Consequence Relation (Rule pair)

A consequence relation, which can be likened to an abstract notion of logical consequence, defines a pattern of reasoning over a given language. It is a mathematical formula that constrains reasoning to specified properties. Consequence relation is defined formally as a set of ordered pairs such that $\{ (\alpha_1,\beta_1),..., (\alpha_n,\beta_n) \}$ with each $\alpha_i,\beta_i \in \mathcal{L}$. It is considered that each ordered pair may contain a set of formulas for the antecedent and a single formula for the consequence. The intuition is that if (α,β) is a consequence relation then $\alpha \vdash \beta$. Any meaningful reasonable notion of inference should follow the properties described by Avron [**?**]:

- **Reflexivity:** $\alpha \vdash \alpha$ for every $\alpha \in \mathcal{L}$
- **Transitivity or Cut:** $\alpha \vdash \gamma \wedge \phi$ and $\phi \wedge \beta \vdash \delta$ then $\alpha \wedge \beta \vdash \gamma \wedge \delta$

3 Kraus, Lehmann and Magidor (KLM) framework

The limitations regarding propositional logic are mainly focused on exceptions within knowledge bases. One way this can be remedied is by denouncing monotonicity as a property. There are many defeasible approaches in the literature and they are all non-monotonic. The preferential approach to non-monotonic reasoning is central to that of the KLM framework. It was first defined by Shahom [**?**] and later extended by Kraus et al [**?**]. The notion of entailment using preferential semantics can be defined by the statement that $\alpha \mid\sim \beta$ which reads as "α preferentially entails β", iff all the preferred models of α also satisfy β. Another way to describe this relationship is that α "typically" holds given β.

3.1 Preferential Consequence Relations

A preferential consequence relation is denoted by $\mid\sim$. As expressed by KLM if $\alpha,\beta \in \mathcal{L}$ then the set of defeasible implications is $\alpha \mid\sim \beta$. The consequence relation, analogous to the classical case is the pair (α, β). Its opposite is $\alpha \not\mid\sim \beta$ means that it is not in the relation. Typically, preferential consequence relations are defined as a set of properties and are represented by inference rules. If the statement above is within the relation then the below statement must be derived and must be in the relation. The consequence relation defined by $\mid\sim$ is a preferential consequence relation iff it satisfies the following properties:

Left Logical Equivalence (LLE):

$$\frac{\models \alpha \leftrightarrow \beta, \alpha \mid\sim \gamma}{\beta \mid\sim \gamma}$$

Right Weakening (RW):

$$\frac{\models \alpha \rightarrow \beta, \gamma \mid\sim \alpha}{\gamma \mid\sim \beta}$$

Reflexivity (Ref):

$$\alpha \mid\sim \alpha$$

And:

$$\frac{\alpha \mid\sim \beta, \alpha \mid\sim \gamma}{\alpha \mid\sim \beta \wedge \gamma}$$

Or:

$$\frac{\alpha \mid\sim \gamma, \beta \mid\sim \gamma}{\alpha \vee \beta \mid\sim \gamma}$$

Cautious Monotonicity (CM):

$$\frac{\alpha \mid\sim \beta, \alpha \mid\sim \gamma}{\alpha \wedge \beta \mid\sim \gamma}$$

The KLM properties or postulates above state for any preferential consequence relation, $\mid\sim$, it must satisfy these properties. Any given pair satisfying the statements above implies that there must be a pair or pairs satisfying the statement below. The first property LLE states that if α and β satisfy each other in the classical notion of entailment then if α defeasibly entails γ then it implies that β will also entail γ. The RW property states that if α implies β in the classical sense then if γ typically entails α, γ must also entail β. The idea of reflexivity is simply the classical notion that α will typically entail itself. The forth property is the conjunctive 'and' which follows that if α entails both β and γ then it must be the case that α

entails the conjunction of the two. Similar to classical propositional logic the 'or' represents that if any two sets of atoms entail γ then their disjunction must also typically entail γ. The final postulate is that of cautious monotonicity which states that if α entails both γ and β without any influence on the other. Cautious monotonicity is the central defeasible property of preferential consequence. CM allows for the addition of premises given a conclusion, it states that inferences that are made can be withdrawn given new information. Thus if α entails γ it can still be the case that α and β entail γ.

3.2 Preferential Interpretations

Preferential semantics are the notion that a set of valuations u are preferred over the set of valuations v. We need to understand what makes an agent selective over specific interpretations. Worlds that are considered more typical or probable depending on the goal will be preferred. Preferential ordering is achieved by the meta notion of states where each state maps to its own interpretations or valuation. We should note that a state particular state may map to many valuations.[**?**]

Definition: A preferential interpretation $\mathcal{P}$ is a tuple $\langle \mathcal{S}, l, \prec \rangle$ with S being a possibly infinite set of states, l:S$\mapsto \mathcal{U}$, and $\prec$ is a strict partial order on S. Given $\alpha \in \mathcal{L}$ and some P, let $[\![\alpha]\!]^{\mathcal{P}} := \{ s \in \mathcal{S}, \mathcal{S} \in \mathcal{P}, l(s) \Vdash \alpha \}$. The definition states that for all states within an interpretation, every valuation from l(s) satisfies α. This statement was proven, in the KLM by contradiction, [**?**] that it is impossible to construct a preferential interpretation such that there are no duplicate states. For all $P = \langle \mathcal{S}, l, \prec \rangle$, a state $s \in \mathcal{S}$ is minimal iff. there is no $s' \in \mathcal{S}$ such that $s' \prec s$. The set of all such s in some P is referred to as $min_{\prec}(P) = \{$ s | there is no s' such that $s' \prec s \}$.

Theorem. $\mid\sim$ *defines a preferential consequence relation if and only if it is the consequence relation defined by a preferential interpretation. In a finite language, every preferential consequence relation is defined by a finite preferential interpretation.*

Reusing the example from earlier we have, in the defeasible context, that,

- Bird $\mid\sim$ Flies
- Penguin $\mid\sim$ Bird
- Penguin $\mid\sim \neg$Flies

We remember that penguins do not exist in the classical context. However, we know that there should be a world in which penguins exist and it should not violate any previous implication about birds. Preferential semantics allow one to construct interpretations where penguins can exist and behave as expected. Let $P = \langle \mathcal{S}, l, \prec \rangle$, such that

- $l(s_1) = bpf$
- $l(s_2) = b\overline{pf}$
- $l(s_3) = \overline{bp}f$
- $l(s_4) = b\overline{p}\overline{f}$
- $l(s_5) = \overline{bp}\overline{f}$
- $l(s_6) = \overline{b}pf$

If we represent the states, such that s is a partial order (antireflexive and transitive). Using P we can test for whether we can satisfy all constraints. $l(s_1)$ satisfies that birds $\mid\sim_P$ fly as the minimal state of bird. Thus the minimal model of bird $\models$ flies. The minimal model for penguin would be s_5 which statisfies birds and therefore penguins will typically imply bird. The last statements minimal model would also be s_5 which does not model flies. Per the theorem above we see that every statement in the set is satisfied by a preferential interpretation, and so there exists a preferential consequence relation, $\mid\sim_{\mathcal{P}}$.

3.3 Ranked Interpretations

Ranked Interpretations can be thought of as multiple tiers of possible states containing any number of interpretations. Ranked interpretations is a subset of preferential ordering. The only difference being that ranked interpretations do not hold duplicate states for specified valuations. If any two states share a valuation then one of the states will be irrelevant. A ranked interpretation can be thought of as a function with a set of valuations as input and a totally ordered set as output (minimal valuations are considered only).

4 Rational Closure

We will not discuss the semantics for rational closure but rather focus on the algorithm for rational closure (RC). Firstly, one must define the BaseRank algorithm. RC is the most conservative of the defeasible reasoning algorithms. It is of the prototypical variety which means it only takes the minimal set of inferences given a Defeasible Knowledge Base, $\mathcal{K}$. Here is the algorithm of BaseRank outlined by Casini et al. [**?**]

Algorithm 1 BaseRank

Input: *A knowledge base* $\mathcal{K}$
Output: *An ordered tuple* $(R_0, ..., R_{n-1}, R_\infty, n)$

```
i := 0;
E_0 := →K;
repeat
    E_{i+1} := {α → β ∈ E_i | E_i ⊨ ¬α};
    R_i := E_i \ E_i + 1;
    i := i + 1;
until E_{i-1} = E_i;
R_∞ := E_{i-1};
if E_{i-1} = ∅ then
    n := i − 1;
else
    n := i;
end if
return (R_0, ..., R_{n-1}, R_∞, n)
```

The above algorithm assigns a ranking to each statement in K based on their exceptionality. Which is any α that makes the knowledge base unsatisfiable. The formula α is said to be exceptional in the set K iff $\{\alpha_1, \dots, \alpha_n\} \cup \{\neg\alpha\}$ is unsatisfiable. $\overrightarrow{\mathcal{K}}$ is the materialisation of $\mathcal{K}$ defined as $\overrightarrow{\mathcal{K}} := \{\alpha \rightarrow \beta \mid \alpha \mid\sim \beta \in \mathcal{K}\}$. Classical statements are given the highest ranking and can never be excluded. Defeasible statement are ranked from most broad to least. Exceptional statements are given the higher ranks as they may violate rules that are more broad. The second part of rational closure is to determine whether a knowledge base entails a defeasible implication. Given an input of a defeasible implication we check it for any conflicts against each rank created from the knowledge base. If we find a

rank that does not conflict with the statement than we can predict it typically holds and in what world it does. If the defeasible implication always conflicts we cannot deduce that it entails the knowledge base. The algorithm is defined as follows:

Algorithm 2 Rational Closure

Input: *A knowledge base* $\mathcal{K}$ and a defeasible implication $\alpha \mid\sim \beta$
Output: **true** if $\mathcal{K} = \mid\approx \alpha \mid\sim \beta$ and **false** otherwise

$(R_0, ..., R_{n-1}, R_\infty, n)$:= BaseRank($\mathcal{K}$);
$i := 0$;
$R := \bigcup_{j=0}^{j<n} R_j$;
while $R_\infty \cup R \models \neg\alpha$ **and** $R \neq \emptyset$ **do**
    $R := R \setminus R_i$;
    $i := i + 1$;
end while
return $R_\infty \cup R \models \alpha \rightarrow \beta$;

4.1 Basic Defeasible Entailment

LM-rationality is defined by Casini et al.[**?**] It follows the KLM postulates. But LM-rationality alone is insufficient for a meaningful form of reasoning.[**?**] Thus, we must consider the extension of inclusion, classical preservation and classic consistency.

- Inclusion: $\forall\ \alpha \mid\sim \beta \in \mathcal{K}, \mathcal{K} \mid\approx \alpha \mid\sim \beta$
- Classic Preservation: $\mathcal{K} \mid\approx \alpha \mid\sim \bot$ iff $\mathcal{K} \mid\approx_P \alpha \mid\sim \bot$
- Classic Consistency: $\mathcal{K} \mid\approx \top \mid\sim \bot$ iff $\mathcal{K} \mid\approx_P \top \mid\sim \bot$

A defeasible entailment relation that satisfies LM-rationality, Inclusion, and Classic Preservation is referred to as a basic defeasible entailment relation. The main derived property is the notion of $\mathcal{K}$-faithfulness:

Definition:. *Given the ranked interpretation* $\mathcal{R}_R^{\mathcal{K}}$ *representing the ranked interpretation corresponding to ranked entailment, any ranked model R of a defeasible knowledge base K is* $\mathcal{K}$*-faithful if the set of valuation in R with non-infinite rank is the same as the set of possible valuations of K. That is,* $\forall\ u \in \mathcal{U}$ *then* $R(u) \neq \infty$.

The DefeasibleEntailment algorithm therefore answers whether a query is entailed according to a particular entailment relation specified by rank. Therefore running DefeasibleEntailment with BaseRank would produce the rational closure output.

Algorithm 3 DefeasibleEntailment

Input: *A knowledge base* $\mathcal{K}$, a $\mathcal{K}$-faithful rank function r, and a defeasible implication $\alpha \mid\sim \beta$
Output: **true** if $\mathcal{K} = \mid\approx \alpha \mid\sim \beta$ and **false** otherwise

$(R_0, ..., R_{n-1}, R_\infty, n)$:= Rank($\mathcal{K}$, r);
$i := 0$;
$R := \bigcup_{j=0}^{j<n} \{R_j\}$;
while $\{R_\infty\} \cup R \models \neg\alpha$ **and** $R \neq \emptyset$ **do**
    $R := R \setminus \{R_i\}$;
    $i := i + 1$;
end while
return $\{R_\infty\} \cup R \models \alpha \rightarrow \beta$;

For the Rank function $[\alpha]$ is defined as: $[\alpha] := \{\beta \mid \alpha \equiv \beta, \beta \in \mathcal{L}\}$

Algorithm 4 Rank

Input: *A knowledge base* $\mathcal{K}$ and a $\mathcal{K}$-faithful rank function r
Output: *An ordered tuple* $(R_0, ..., R_{n-1}, R_\infty, n)$

$iR_\infty := \neg(\bigvee_{r([\alpha])=\infty}[\alpha])$;
$n := \max\{i \in \mathbb{N} \mid \exists\alpha \in \mathcal{L}\ s.t.\ r(\alpha) = i\}$;
if $n = 0$ **then**
    $R_0 := \top$;
    $n := 1$;
else
    for $i := 0$ to n - 1 **do**
        $R_i \equiv_{def} \neg(\bigvee_{r([\alpha])=i+1}[\alpha])$
    end for
end if
return $(R_0, ..., R_{n-1}, R_\infty, n)$

The output of the Rank algorithm is a tuple of formulas logically equivalent to the material counterparts to the defeasible implications given as input. The DefeasibleEntailment algorithm is used to determine whether a given defeasible implication is entailed by $\mathcal{K}$.

4.2 Lexicographic Closure

Lexicographic closure is the presumptive reasoning approach as compared to the prototypical approach of RC.[**?**] Presumption of typicality is thought to be a stronger assumption of monotonicity in the absence of information that would otherwise make us retract inferences. Rational entailment does not give us a strict sense of what information to derive only that new information must not contradict existing knowledge. Essentially presumptive reasoning prefers to derive inferences that are more typical given the knowledge base whilst there being no strict directive to derive information. The concern regarding the presumptive approach is that some derivable inferences are valid but some are not and may be superfluous to the knowledge base. It could also be that various inferences could clash based off defeasible implications, in this case the more specific antecedent is preferred.

Example: Given $K = \{ p \rightarrow b, b \mid\sim f, p \mid\sim \neg f, r \rightarrow b, b \mid\sim w \}$.

The prototypical approach here would not conclude that $p \mid\sim w$ because there is no relation between p and w. The presumptive approach on the other hand would be able to derive this inference because p is typically b and since b typically infers w, thus it is intuitive that $p \mid\sim w$. It is also plausible given new information that this derivation may not hold but given our knowledge base this is the only preferred inference with the presumptive approach.

Defintion *Given a defeasible knowledge base K, the order of a knowledge base, k, is the maximum base rank assigned to a formula in the knowledge base, not including* ∞*, such that* $k^K := max\{\ br(\alpha) \mid \alpha \mid\sim \beta \in K,\ br(\alpha) \in \mathcal{N}\}$

To order such subset we clarify a notion of set of defeasible implication as more "serious" than another. It is defined by the following two metrics.[**?**]

- the size of the set: if the cardinality is larger then it is more serious and if it is smaller than it is less.
- the specificity of the elements: it is less serious to violate less specific than more specific ones. This is informed by the base rank of the statement.

The theorem presented by Lehmann is as follows:
Theorem. *Given a defeasible knowledge base $\mathcal{K}$, and a defeasible implication $\alpha \mid\sim \beta$, it is the case that $\alpha \mid\sim \beta$ is in the lexicographic closure of $\mathcal{K}$, denoted $\mathcal{K} \mid\approx_{\mathrm{LC}} \alpha \mid\sim \beta$, if and only if for any basis $B \subseteq \mathcal{K}$ of α, $\overrightarrow{B} \cup \{\alpha\} \models \beta$.*
The seriousness of ordering can be used to construct ranked interpretation of K that correspond to lexicographic closure. Thus lexicographic closure is said to be LM-rational [**?**]. The rank function said to generate lexicographic closure is as follows: *The lexicographic rank function, $r^{LC}_{\mathcal{K}}$, with respect to a knowledge base $\mathcal{K}$, is defined as $r^{LC}_{\mathcal{K}}(\alpha) := min\{R^{LC}_{\mathcal{K}}(v) \mid v \in \hat{\alpha}\}$*
Therefore we can compute lexicographic closure with the same algorithms that compute basic defeasible entailment.

4.3 System W

System W is an inductive inference operator that exhibits properties that extend rational closure. The idea is to lift the limitation so that it can impose some worlds to be infeasible [**?**]. Komo and Beierle presented System W, which can capture and extend both c-inference and system Z inference [**?**]. It is noted that lexicographic inference, modified to be an inductive inference operator, can capture System W^+.[**?**] System W is defined using both the inclusion-maximal ordered partition OP(R) and the information on the set of falsified conditionals. [**?**] System W tracks the conditionals at each rank and falsifies them by a strict subset of a strict partial order $<^W_R$. This is referred to as system W inference and is defined by Komo and Beirele as follows:
Definition: Let $\mathcal{R}$ be a knowledge base and let A, B be formulas. Ω is denoted as the set of all possible worlds. Then B is a system W inference from A (in the context of $\mathcal{R}$), denoted

$$\mathrm{A} \mid\sim^{\mathrm{w}}_{\mathcal{R}} \mathrm{B} \text{ iff } \forall\, \omega' \in \Omega_{A\overline{B}}\ \exists\, \omega \in \Omega_{AB}\ \omega <^{\mathrm{w}}_{\mathcal{R}} \omega.$$

The preference relation of a preferred world ω is preferred if it falsifies less conditionals than ω' and ω' falsifies at least the same conditionals as ω. The algorithm for system W inference has been implemented by Beierle et al. [**?**] as Partial MaxSAT problems. The model for system W follows that given an input of the belief base Δ and sets of formulas A and B, the output would be the truth value of if $A \mid\sim^W_\Delta$ B holds.

5 DHCE and ECCRS

The Defeasible Horn Classifier with Exceptions (DHCE) is a symbolic classifier designed to provide interpretability. It utilizes a "normally...unless..." pattern for classification tasks. In contrast to the Single Boolean Formula (SBF) approach, DHCE distinguishes between defaults and exceptions in classification scenarios.[**?**] SBF approaches lack a notion of rule priority, which limits their interpretability compared to DHCE.[**?**] Exception Closed Conjunctive Rule Sets (ECCRS) represent another classification method that offers clear and distinct justification and explanation. Each rule in ECCRS is structured as an if-then statement, enabling human readability. Predictions can be justified by referencing the specific if clause that triggered the outcome. For each instance, a set of encoded rules indicates which rules were applied and which were overridden. This structure enables precise justification to accompany each prediction.

ECCRS assumes a booleanised feature set of literals that are given based of a specific instance. ECCRS rule is a pair (B, l) where

- B is the conjunction of signed literals (the rule body)
- l is the class label either { 0,1 }

The rule (B, l) is only applicable if an instance satisfies all possible literals of B. The key structural relation of ECCRS is that if we have rule sets B and C then we have the following possibilities:

- B and C are said to be compatible if there are no such contradictory literals
- if B $\subset$ C then C has more literals and is said to be more specific
- B and C are comparable if B $\subset$ C or C $\subset$ B.

5.1 Exception Closure

Rulesets obtained by ECCRS are constrained such that any opposite labeled rules cannot conflict in any unstructured way. When two rules are applicable in the same instance and neither is more specific then a tie break is needed.

5.1.1 Relaxed (local) closure. A ruleset R is said to be relaxed exception closed in D if opposite labeled rules are required to be structurally comparable - when they overlap in D. Whenever a default and an exception can be imposed on the same row then the exception must be visibly more specific. Relaxed closure does not force them into co applying.

5.1.2 Strict (global) closure. R is said to be strictly exception closed if it is made comparable on the entire booleanised feature space. For all (B,l) and (C,1 - l) in R, their rule bodies must be comparable by strict inclusion. B $\in$ C or C $\in$ B. Strict global closure prohibits any opposite label conflicts in the feature space. The idea is that ECCRS will only be compatible with KLM translation if their are no conflicts. The RC reasoner is only defined of complete feature patterns.

5.2 Most Specific Wins

Definition: Let F be an instance pattern such that $A(F) :=$ { (B,l) $\in$ R: F $\models$ B } is the applicable set of rules.

Definition: The inclusion-maximal set of applicable rules is defined as: $M(F) :=$ { (B,l) $\in A(F) :$ (C,l') $\notin A(F)$, B $\subset$ C}

Most specific Wins is therefore defined as follows with respect to instance pattern F.

- MSW_R(F) = l if $M(F) \neq \emptyset$, $\forall$ M(F) have label l.
- MSW_R(F) = $\perp$ if $M(F) = \emptyset$
- MSW_R(F) = $\perp$ if $M(F)$ contains both labels

Example: Given a set of atoms p_1 .. p_5. We can a booleanised feature set as follow:

$$r_1 : (p_1) \Rightarrow 1$$
$$r_2 : (p_1 \wedge p_4) \Rightarrow 0$$
$$r_3 : (p_1 \wedge p_4 \wedge p_5) \Rightarrow 1$$
$$r_4 : (p_2 \wedge p_3) \Rightarrow 1$$
$$r_5 : (p_2 \wedge p_3 \wedge p_4) \Rightarrow 0$$

If we define a feature pattern F := $p_1 \wedge \neg p_2 \wedge \neg p_3 \wedge p_4 \wedge p_5$. We see that we have an exception to an exception such that M(F) = $\{r_3\}$ and $MSW_R = 1$

- r_1 applies
- r_2 overrides r_1
- r_3 overrides r_2

Example 2: We define a feature pattern F := $p_1 \wedge \neg p_2 \wedge \neg p_3 \wedge p_4 \wedge \neg p_5$. Since p_5 is false we see that r_2 is the most specific that applies thus $MSW_R = 0$.
Example 3: We define a feature pattern F := $\neg p_1 \wedge p_2 \wedge p_3 \wedge \neg p_4 \wedge \neg p_5$. The only applicable rule is r_4. Thus $MSW_R = 1$.
Example 4: If no rule pattern is uncovered it must be the case that no rule body is satisfied. Thus MSW_R (F)= $\perp$.
Considering these four outcomes we can safely conclude that given atoms of any kind there is sufficient justification for the prediction. The most specific wins principle allows one to see exactly which rule justified the conclusion.

6 Discussion

Rulesets that are deduced, such as those from ECCRS is a new approach to extract justifiable explanations from predictive datasets. Symbolic classifiers are well established and can benefit from existing defeasible knowledge frameworks and algorithms, which rigorously justify entailment relations. Symbolic classifiers could leverage knowledge representation as a tool to confirm and justify concrete logical consequences. The subsequent project would lend itself to creating a link between ECCRS and defeasible knowledge bases. Thereby allowing the use of various known algorithms for defeasible entailment and ranking. Thereby, cross-examining ECCRS rulesets with defeasible entailment. The structural gap with this however is that ECCRS rulesets will need in a comparable structure to ECCRS and may be phrased at a higher abstraction level. Noticeable, rational closure provide defeasible entailment at the object level [**?**] and rulesets from ECCRS does not. The alignment theorem from most specific wins to RC should although for RC queries to match with the predictions from ECCRS. Explanation traces for RC are only concerned with the query result, does it entail or not. Whereas, for explanation traces regarding ECCRS we can specify each exception-to-exception as the trace.

7 Conclusions

In this literature review, we have explored the foundations of classical propositional logic and its nonmonotonic extension in defeasible reasoning. We provide some of the background needed for the implementation of specific defeasible algorithms. The background of their semantics will guide the translation between ECCRS rulesets to the KLM framework, which would allow for further exploration with regard to basic defeasible entailment relations. The algorithms presented will be essential in producing a toolkit that verifies ECCRS rulesets. Lastly, a suggestion for research would be to allow for extensions that align knowledge bases with the constraints imposed by either lexicographic closure and System W.

References